MATH-540 HW #5. LS.

Problem 1. (Z&T Ch. 3. Pr. 5.1)

Consider the sol-n to the <u>conserv-n law</u> eq-n

(1) $$z = f(x - a(z)y),$$

(a) Show that the sol-n, (1), is a function.

(b) Determine z_x & z_y from (1).

(c) Verify that $z(x,y)$, as defined by (1), satisfies the conservation law eq-n:

(2) $$a(z)\cdot z_x + z_y = 0$$ and the Initial Curve

(3) $$z(x,0) = f(x).$$

Sol-n:

(a) To show the sol-n given by (1) is a fun-n $z = z(x,y)$, we can use Implicit Function Theorem:

Implicit Function Theorem:

Let $F(x,y,z) \in C^1(\mathbb{R}^3)$.

If $\frac{\partial F}{\partial z}(x_0, y_0, z_0) \neq 0$, then $F(x,y,z) = C$ "can be solved for z" in terms of x & y in a <u>nbd</u> of (x_0, y_0, z_0), i.e. $z = f(x,y)$, and

(4) $$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \& \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}.$$

To show that $\exists$ a sol-n $z = z(x,y)$ given by (1), let's form the implicit f-n as follows:

(5) $$F(x,y,z) = z - f(x - a(z)\cdot y) = 0.$$

Then,

using "Chain rule":

$$\frac{\partial F}{\partial z}\Big| = \frac{\partial}{\partial z}\left(z - f(x - a(z)y)\right) = 1 - f'(x - a(z)\cdot y)\cdot\left[-a'(z)\cdot y\right]$$

"should be"

(6) $$\Rightarrow \frac{\partial F}{\partial z}\Big|_{(x,y,z)} = 1 - f'(x - a(z)\cdot y)\cdot\left[-a'(z)\cdot y\right] \neq 0 \Rightarrow \boxed{F_z > 0 \text{ or } F_z < 0}$$

Considering the IC in (3), $\boxed{y=0}$, replacing this into (5), yields

$$\frac{\partial F}{\partial z}\Big|_{(x,0,z)} = 1 - f'(x - a(z)\cdot 0)\cdot\left[-a'(z)\cdot 0\right] = 1 - 0 = 1 > 0$$

$$\frac{\partial F}{\partial z}\Big|_{(x,0,z)} = 1 > 0$$ - in a <u>nbd. of IC</u>!

We should consider $\boxed{F_z > 0}$ case <u>only</u>, i.e. our implicit sol-n <u>exists</u> as a function of x, y, i.e. $z = z(x,y)$, as long as

(7) $$\frac{\partial F}{\partial z}\Big|_{(x,y,z)} = 1 + f'(x - a(z)\cdot y)\cdot a'(z)\cdot y > 0$$

So for pts that satisfy (7), the sol-n given by (1) is a function $z = z(x,y)$.

$$z_x = \frac{F_x}{F_z} = \frac{f'(x - a(z)\cdot y)}{1 + f'(x - a(z)y)\cdot a'(z)\cdot y}, \quad \text{as } F_z \to 0, \quad z_x \to \pm\infty.$$

(b) $z_x = ?$ $z_y = ?$ using (1):

(1) $$z(x,y) = f(x - a(z)\cdot y) \quad \left\{ \frac{\partial(\cdot)}{\partial x} \text{ of BS:} \right.$$

$$\underbrace{\frac{\partial}{\partial x} z(x,y)}_{z_x} = \frac{d}{dx} f(x - a(z)\cdot y) \quad \Rightarrow$$

$$z_x = f'(x - a(z)\cdot y)\cdot \frac{\partial}{\partial x}(x - a(z)\cdot y) =$$

$$= f'(x - a(z)\cdot y)\cdot \left(1 - \frac{\partial}{\partial x}[a(z)\cdot y]\right) =$$

$$= f'(x - a(z)\cdot y)\cdot (1 - a'(z)\cdot z_x \cdot y). \quad \Rightarrow$$

$$z_x = f'(x - a(z)\cdot y)\cdot [1 - a'(z)\cdot z_x \cdot y] \quad \Rightarrow$$

(8) $$\boxed{z_x = \frac{f'(x - a(z)\cdot y)}{1 + a'(z)\cdot y\cdot f'(x - a(z)\cdot y)}}$$, and

for z_y, to avoid the calculations, we can use IFT:

$$z_y = -\frac{F_y^{\text{by (5)}}}{F_z^{\text{by (6)}}} = -\frac{-f'(x - a(z)\cdot y)\cdot[-a(z)]}{1 + f'(x - a(z)\cdot y)\cdot a'(z)\cdot y} = \frac{-f'(x - a(z)y)\cdot a(z)}{1 + f'(x - a(z)y)\cdot a'(z)\cdot y}$$

(9) $$\boxed{z_y = \frac{-f'(x - a(z)y)\cdot a(z)}{1 + f'(x - a(z)\cdot y)\cdot a'(z)\cdot y}}.$$

(c) Verifying the sol-n:

First, let me check the IC, & then the eq-n:

IC: $z(x,0) \overset{(1)}{\underset{y=0}{=}} f(x - a(z)\cdot 0) = f(x)$ ✓

Eq-n: $a(z)\cdot z_x + z_y =$ (Using the results in (b))

$$= a(z)\cdot\left[\frac{f'(x-a(z)\cdot y)}{1+f'(x-a(z)y)\cdot a'(z)\cdot y}\right] + \left[\frac{-a(z)\cdot f'(x-a(z)\cdot y)}{1+f'(x-a(z)\cdot y)\cdot a'(z)\cdot y}\right]$$

$\equiv 0$ ✓

Hence, the f-n, defined in (1), is a sol-n for (1) + (2).

Problem 2. (Z&T, Ch 3, Pr. 5.2)

Consider the conservation law:

(1) $a(z)\cdot z_x + z_y = 0$ with IC

(2) $z(x,0) = f(x)$

Under the condition that $a(z) = a$ is a constant.

Show that no shocks develop.

Sol-n: There are 2 ways of showing when shocks do (not) develop:

① Algebraically (Using the IFT):

(3) $F_z = 1 + f'(x - a(z)y)\cdot a'(z)y > 0$.

But now since $a(z) = a$ & $a'(z) = 0$, (3) becomes

$F_z = 1 + f'(\ldots)\cdot 0\cdot y = 1 > 0$ ✓. $\Rightarrow$

There's no SHOCKS.

Geometrically, we could try drawing characteristics $\langle x(s), y(s)\rangle$ by

(4) $\boxed{x - x_0 = a(z_0)\cdot y}$, where $x_0 = \tau$,

$\Downarrow$

$\boxed{y = \frac{1}{a(z_0)}(x - x_0)} \xrightarrow[\& \ a \neq 0]{a(z_0) = a} y = \frac{1}{a}(x - x_0) \Rightarrow$

✓

If $a \equiv 0$, then $x = x_0$, the char-cs never meet! ▨

Prob. 3. Z&T. Ch 3, Pr. 5.3.

Consider the conservation law:

(1) $a(z)\, z_x + z_y = 0$

(2) $z(x,0) = f(x)$

where $f(x) = k$ is a const. Do shocks develop?

Sol-n. Given (1) + (2), there are 2 ways.

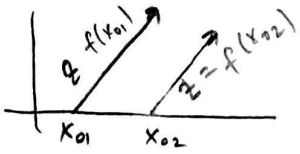
(i) Algebraically:

(3) $F_z = 1 + \underbrace{f'(x - a(z)y)}_{k' = 0}\cdot a'(z)y = 1 + 0\cdot a'(z)y = 1 > 0.$ ✓ $\Rightarrow$

No shocks occur!

(ii) Geometrically:

(4) $\boxed{x - x_0 = a(\underbrace{f(x_0)}_{k})\cdot y} \Rightarrow \boxed{y = \frac{1}{a(k)}(x - x_0)} \Rightarrow$



If $a(k) \equiv 0$, then, $x = x_0$, characteristics <u>never</u> cross, otherwise, $a(k) = \text{const}$, all of the characteristics are parallel, hence <u>never</u> cross again! $\Rightarrow$ No shocks! ▨.

Problem 4. Z&T, Ch 3, Pr. 5.5.

Consider

(1) $z\cdot z_x + z_y = 0$

(2) $z(x,0) = x.$

(a) Find the explic. sol-n

(b). Determine & sketch the charact-cs

(c) Do shocks develop for $y > 0$?

(a)

Sol-n: We know that given the eq-n (1) + (2) where

(3) $a(z) = z, \quad f(x) = x \in C^1(\Omega)$

the general sol-n is

(4) $$\boxed{z = f(x - a(z)\cdot y)}.$$

Thus,

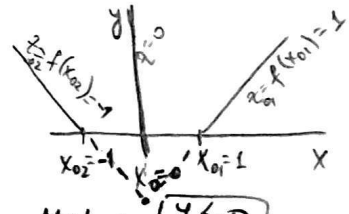
(5) $$z = f(x - zy) = x - zy \Rightarrow \boxed{z = \frac{x}{1+y}}$$

(6) Charact-cs:

$$x - x_0 = a(z_0)\cdot y \Rightarrow y = \frac{x - x_0}{a(z_0)} = \frac{x - x_0}{z_0} = \frac{x - x_0}{f(x_0)} = \frac{x - x_0}{x_0}$$

$$\Rightarrow \boxed{y = \frac{1}{x_0}x - 1} \text{ (lines)}$$

x_0	0	1	-1
$y(x)$	$x=0$	$y = x-1$	$y = -x-1$



Meet at $\boxed{y < 0}$

(c) Shocks:

(i) Algebraically,

$$F_z = 1 + f'(x - a(z)\cdot y)\cdot a'(z)\, y = 1 + [1]\Big|_{x - zy} \cdot z'\cdot y = 1 + (1)\cdot(1)\cdot y > 0$$

when $y > 0$, so <u>no shocks</u>!

(ii) graphically,

we can see that the intersection of characteristics is @ $y < 0$,

so for $y > 0$, there's <u>no shocks</u>!

Problem 5. Z&T, Ch. 3, Pr. 5.6

(a) Consider the IV pr-m

(1) $z^2 \cdot z_x + z_y = 0$

(2) $z(x, 0) = x.$

Derive the sol-n

(3) $$z(x,y) = \begin{cases} -1 + \sqrt{1 + 4xy}, & y > 0 \\ x, & y = 0 \end{cases}.$$

(b) Show that this sol-n in (3) is at <u>least</u> C^0 (that is recall $C^0 = \{f(x) : f(x) \text{ is continuous in its domain}\}$) by showing $\lim_{y \to 0} z(x,y) = x$.

Because recall the def-n of continuity:
① $f(a)$ defined
② $\lim_{x \to a} f(x)$ exists ③ $\lim_{x \to a} f(x) = f(a)$

(4)

To show the sol-n in (3) is cont-s in it <u>whole</u> domain, we know sol-n is cont-s for $y \neq 0$ for $y > 0$ part. The only problem might be at $y = 0$. By IC, we have the sol-n f-n value at $y = 0$, i.e $z(x,0) = x$. So for continuity of the sol-n fun-n is then defined by $\lim_{y \to 0} z(x,y) = z(x,0) = x$. That's why

(5) Show $\boxed{\lim_{y \to 0} z(x,y) = x.}$

(Hint: By Taylors series, show that $\sqrt{1+u} \simeq 1 + \frac{1}{2}u$ when u is small)

(c) When do shocks develop?

Sol-n: (9) (1) + (2) is a conservation law with

(6) $a(z) = z^2$ & $f(x) = x \in C^1$, then we know the sol-n

(7) $z = f(x - a(z)y) = f(x - z^2 y) = x - z^2 y$. $\Rightarrow$

(7) $\boxed{z = x - z^2 y}$ $\Rightarrow$ $y \cdot z^2 + z - x = 0$

(8) $$z_1 = \frac{-1 - \sqrt{1+4xy}}{2y} \quad \& \quad z_2 = \frac{-1 + \sqrt{1+4xy}}{2y}$$

Using the IC for both z_1 & z_2 to determine which one to pick:

(9) $$z_1(x,0) = \frac{-1 - \sqrt{1+4x \cdot 0}}{2 \cdot 0} = \frac{-1-1}{2 \cdot 0} = \frac{-1}{0} \neq x$$

(10) $$z_2(x,0) = \frac{-1 + \sqrt{1+4x \cdot 0}}{2 \cdot 0} = \frac{-1+1}{0} = \frac{0}{0} \text{ (indeterminate form)}$$

Now for z_1, it is <u>impossible</u> <u>to be a sol-n</u>! But for z_2 there's a <u>chance</u>!

However, since the given problem satisfies the requirements of existance & uniqueness thm ($P = z^2$, $Q = 1$, $R = 0$, $f(x) = x \in C^1(\cdot)$ & $\begin{vmatrix} P|_{IC} & Q|_{IC} \\ x_0' & y_0' \end{vmatrix} \neq 0$) we know that there exist a unique sol-n. So, if we exclude the $z_1 = (1 - \sqrt{\cdot})/(2y)$ sol-n case, we are <u>only</u> left with one option

(11) $$z(x,y) = \frac{-1 + \sqrt{1 + 4xy}}{2y}, \quad y > 0$$

thus matches the suggested sol-n in (3). $\blacksquare$

(b)

$\lim_{y \to 0} z(x,y) = z(x,0) = x$, Check this:

$$\lim_{y \to 0} \frac{-1 + \sqrt{1+4xy}}{2y} \underset{\text{series for } \sqrt{1+x} = 1 + \frac{x}{2} + \dots}{\overset{\text{by the hint Maclaurin}}{=}} \lim_{y \to 0} \frac{-1 + 1 + \frac{1}{2}(4xy)}{2y}$$

$$= \lim_{y \to 0} \frac{2xy}{2y} = \lim_{y \to 0} x = x$$

hence $\lim_{y \to 0} z(x,y) = z(x,0)$ this sol-n in (3) is at least in C^0. $\blacksquare$

(c) When do shocks develop?

Using the shocks cond-n,

$$F_z = 1 + f'(x - a(z)y) \cdot a'(z)y = 1 + (1) \cdot 2zy = 0$$

$$\boxed{z = -\frac{1}{2y}}$$. Finding the $\langle x(s), y(s) \rangle$ characteristics relation using that, plug that into the derived sol-n for $y > 0$ part.

So algebraically

(12) $$\frac{\sqrt{1+4xy}-1}{2y} = -\frac{1}{2y} \Rightarrow \boxed{y = -\frac{1}{4x}}$$

So the case in (12) is the shock occurrance case! ▨

Problem 6. ZRT, Ch. 3. Pr 6.1. Consider the ↓ traffic pr-m

(1) $$\frac{\partial d}{\partial t} + \underbrace{c(1-2d)}_{a(d)} \frac{\partial d}{\partial x} = 0$$

(2) $$d(x,0) = f(x) = \begin{cases} 0, & x \le 0 \\ x/L, & 0 \le x \le L \\ 1 & L \le x \end{cases}$$

(a) When a shock happens?

(b) Derive the sol-n:

(3) $$d = \begin{cases} 0, & x \le ct \quad \text{left reg-n} \\ \frac{x-ct}{L-2ct}, & 0 < \frac{L(x-ct)}{L-2ct} < L, \text{ middle } \Delta \text{ reg-n} \\ 1, & x \ge L-ct \quad \text{right reg-n} \end{cases}$$

(c) Sketch a few characteristics.

Sol-n: (a) Using the shock cond-n, for the sol-n $d = d(x,t)$

$$F_z = 1 + f'(x - a(d)t) \cdot a'(d) y = 1 - 2ct f'(x - ct(1-2d)) =$$

(4) $$= 1 - 2ct \cdot \left[\begin{cases} 0 & x - ct(1-2d) < 0 \\ \frac{1}{L} & 0 < x - ct(1-2d) < L \\ 0 & x - ct(1-2d) > L \end{cases} \right], \text{ so}$$

when $x - ct(1-2d) < 0$ or $x - ct(1-2d) > L$, $F_z = 1 - 2ct \cdot 0 = 1 \neq 0$

there's no shocks.

So we expect shocks to happen when $0 < x - ct(1-2d) < L$ &

(5) $F_2 = 1 - 2ct \cdot \left[\frac{1}{L}\right] = 0 \Rightarrow \boxed{ct = \frac{L}{2}}$ (5) (when middle cond-n is $= 0$).

(b) Sol-n

Recall, for conservation law prob-s, the sol-n

$$d = f(x - ct(1-2d)).$$

(6) • Left side: $\boxed{d = 0}$ (6), this happens when the argument of f, $x - ct(1-2d) \leq 0 \Rightarrow$ (6)

$$x - ct(1-0) = 0 \Rightarrow \boxed{x - ct \leq 0}$$

(7) • Right side: $d = 1$ (7), when the argument of f, by (2)

$$x - ct(1 - 2\underbrace{d}_{1}) \geq L$$

$$x + ct \geq L \quad (7)$$

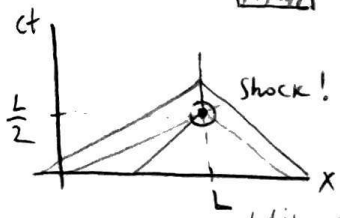
• Middle side:

$$d = \frac{x - ct(1-2d)}{L} \quad \text{when} \quad 0 < x - ct(1-2d) \leq L \xrightarrow[\text{for } d]{\text{Solving it}}$$

(8) $$d = \frac{x - ct}{L - 2ct} \quad \text{when} \quad 0 \leq x - ct\left[1 - 2\left(\frac{x - ct}{L - 2ct}\right)\right] \leq L$$

(c) Characteristics: $x - x_0 = a(z_0) \cdot y \Rightarrow$

$$x - c(1-2d_0)t = x_0$$



• $x_0 \leq 0$, $d_0(x_0) = 0$, $ct = x - x_0$ (lines with slope 1)

• $x_0 \geq L$, $d_0(x_0) = 1$, $x - ct + 2ct = x + ct = x_0 \Rightarrow ct = x_0 - x$ (lines with slop -1)

• $0 < x_0 < L$, $d_0(x_0) = \frac{x_0}{L}$, $ct = \frac{L(x - x_0)}{L - 2x_0}$ (lines with slope $\frac{L}{L - 2x_0}$).